

1 Experimental design

Table 1: Variables, methods, and justifications

Independent variable	Method of how this variable will be changed	Justification: Why this range, interval or frequency was chosen?
Distance between Slit and screen	Forthe long range,there will be three objects the pattern will be projected on,the object chosen is a graphed piece of paper on a clipboard, suspended hanging on a table. For each distance, the source will move between 3 distances, and once the data for each table is collected, it will be moved away to revealthe nexttable. For the shortrange data, the projected screen will move on a wheel on tracks.	This range of data will provide us justification for not only the short range, but also long range to show the validity ofthe double slit equation over a range of distances.
Dependent Variable	Method of how this variable will be measured	Justification: How many trials? How will precise and sufficient data be collected?
The distance between adjacent-fringes on the projected screen	I willtake a high resolution image with my phone and use perspective correction on the image and use software solutions such as imageJto find the average distance between each fringe	One picture will be taken for each distance.But multiple takes on fringe spacing will be recorded and the average will be found.
Control Variable	Potential impacts of this variable on the dependent variable/data	How it will be controlled and kept constant
The same laser and slit same method is used to measure the distances		

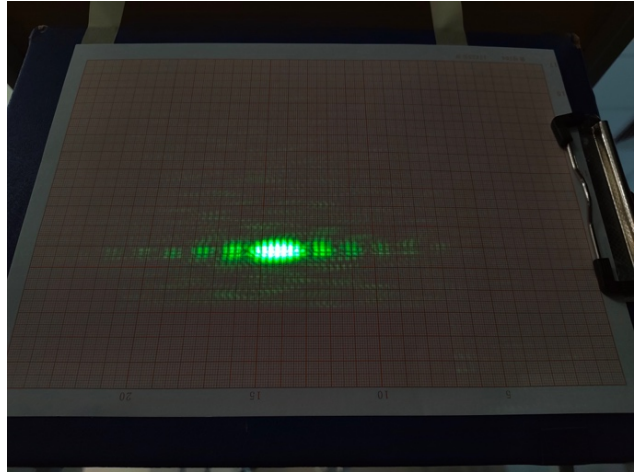
2 Data analysis

Table 2: Qualitative observations and interpretations

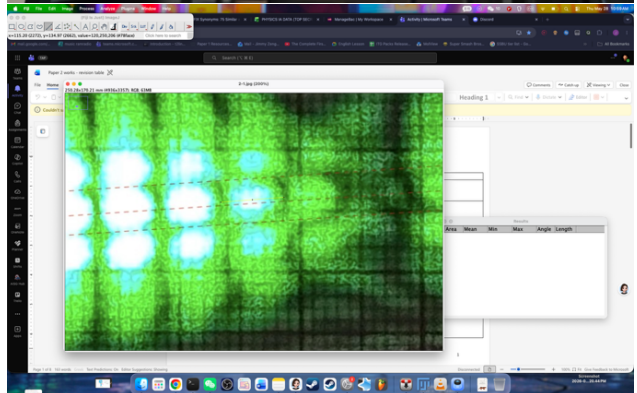
Qualitative observations	Interpretations/Photos
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I don't recall having taped the paper onto the surface of the clipboard for maximum flatness of the surface. The paper is clipped on the right side but not the left

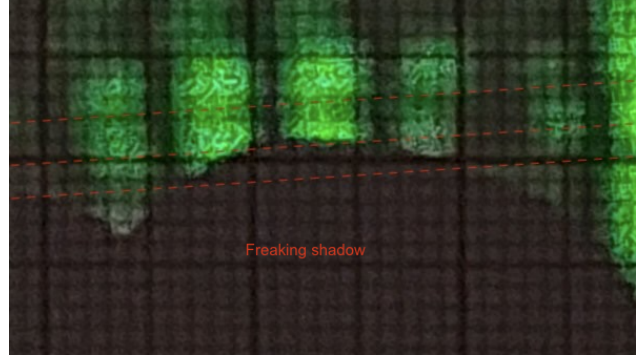
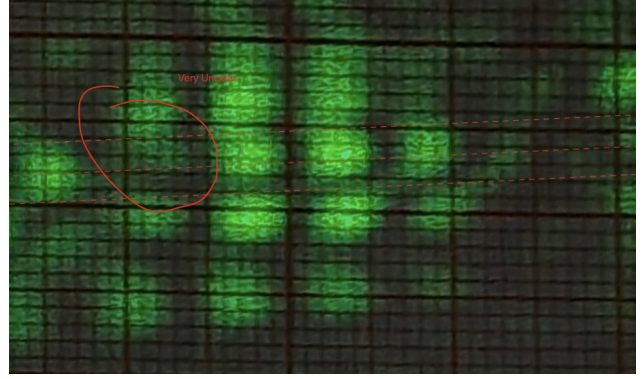
This means when we do the perspective shifts it might have slight distortions



Some are really slanted for some reason. I ended up correcting it by drawing lines with respect to the direction of the slant and then measuring with that angle



Some of the fringes are too unclear to be seen
I have noted them in my data when there were
issues measuring



There are shadows sometimes in the data. This is
also noted

3 Raw and processed measurements

Table 3: Measurements for data sets 1-1, 1-2, and 1-3

	1-1		1-2		1-3	
Pixels per cm square side length (pixels/cm). Used as a reference of size.	334.005	332.667	149.678	150.251	149.169	150.667
Pixels per cm square side length (pixels/cm). Used as a reference of size.	323.488	329.334	151.627	151.503	150.334	153.004
Pixels per cm square side length (pixels/cm). Used as a reference of size.	332.669	332.003	154.002	149.231	150.17	149.333
Mean	330.694		151.049		150.4461667	
Measurement of paper graph square side length after setting the mean (above) as reference	1.036	1.005	1.004	0.949	1.000	0.989
Measurement of paper graph square side length after setting the mean (above) as reference	1.013	0.953	1.081	1.01	0.975	0.972
Measurement of paper graph square side length after setting the mean (above) as reference	0.983		0.977		1.022	

Central Maxima (mm). Measured as the distance from the left/right edge of one bright fringe to the left/right edge of an adjacent bright fringe	Left edge	Right Edge	Left edge	Right Edge	Left edge	Right Edge
Central Maxima (mm). Measured as the distance from the left/right edge of one bright fringe to the left/right edge of an adjacent bright fringe	3.044	2.822	3.078	3.509	4.254	4.188
Central Maxima (mm). Measured as the distance from the left/right edge of one bright fringe to the left/right edge of an adjacent bright fringe	2.903	2.984	3.443	3.443	4.52	4.143
Central Maxima (mm). Measured as the distance from the left/right edge of one bright fringe to the left/right edge of an adjacent bright fringe	3.014	2.923	3.608	3.575	4.032	4.055
Central Maxima (mm). Measured as the distance from the left/right edge of one bright fringe to the left/right edge of an adjacent bright fringe	2.863	2.903	3.542	3.525	4.165	4.143
Central Maxima (mm). Measured as the distance from the left/right edge of one bright fringe to the left/right edge of an adjacent bright fringe	2.963	2.903	3.542	3.426	4.099	4.099
Central Maxima (mm). Measured as the distance from the left/right edge of one bright fringe to the left/right edge of an adjacent bright fringe	2.974	2.812	3.492	3.525	4.143	4.365
Central Maxima (mm). Measured as the distance from the left/right edge of one bright fringe to the left/right edge of an adjacent bright fringe			3.343	3.525	4.055	4.032
Central Maxima (mm). Measured as the distance from the left/right edge of one bright fringe to the left/right edge of an adjacent bright fringe			3.542			
First Maxima (mm)	3.014	2.792	3.343	3.409	3.9	3.966
First Maxima (mm)	3.105	3.095	3.492	3.36	4.365	4.099
First Maxima (mm)	3.064	3.054	3.542	3.227	4.431	3.811
First Maxima (mm)	2.974	3.155	3.492	3.906	4.232	4.055
Second Maxima	2.933		3.426	3.062	4.032	3.877
Second Maxima	2.979		3.509	3.625	4.165	3.988

Second Maxima	2.903	3.294	3.707	4.032	4.032
(Data deemed difficult to measure consistently for the first one)	2.767	3.459	3.045	4.188	3.955
(Data deemed difficult to measure consistently for the first one)	3.22	3.261	3.327	4.276	4.154
(Data deemed difficult to measure consistently for the first one)	2.812	3.095	3.757	4.121	4.104
Average fringe displacement (mm)	2.96805	3.4416		4.119882353	
Expected/Calculated Value (mm)	2.9295	3.521		4.112	
Distance between laser and screen (cm)	247.8	297.8		347.8	
uncertainty of distance between adjacent bright fringes (mm)	± 0.2265	± 0.4305		± 0.3545	

Table 4: Measurements for data sets 2-1, 2-2, 2-3, and 3-1

2-1		2-2		2-3		3-1	
198.668	197.333	247.335	249.668	236.513	237.334	162	162.507
196.667	198.669	248	247.337	235.675	237.51	162.501	161.757
196.667	195.336	248.002	247	225.281	229.847	157.009	161.174
197.223		247.89		233.693		161.158	
1.014	1.031	1.059	0.968	0.999	0.956	1.003	1.034
1.014	1.048	1.019	0.98	1.002	1.027	1.016	1.081
0.98		0.989		1.038		1.024	
Left edge	Right Edge	Left edge	Right Edge	Left edge	Right Edge	Left edge	Right Edge
4.91	4.322	6.102	5.866	6.994	6.262	6.144	6.408
5.542	5.137	5.922	5.883	6.39	6.342	6.227	6.557
5.306	5.316	6.099	5.74	6.524	6.393	6.455	6.186
5.129	5.219	5.839	5.901	6.436	6.601	6.599	6.517
5.287	5.215	6.042	6.042	6.479	6.61	6.683	6.417
5.479	5.413	5.987	5.9	6.302	6.308	6.372	6.31
5.186	5.262	5.984		6.479	6.732	6.537	6.682
5.108	5.371						
5.08	5.026	5.781	5.696	6.226	6.516	6.081	6.175
5.184	5.434	6.066	6.061	6.664	6.601	6.765	6.484
5.592	5.339	5.957	5.826	6.863	6.617	6.548	6.298
6.092	5.643	5.751	6.287	6.644	6.702	6.795	6.578
bad data		only clear data is extracted	only clear data is extracted			surprisingly consistent	surprisingly consistent
4.928		5.536	5.018	6.025	5.866	6.488	6.484
5.192		5.658	5.462	6.366	6.253	6.454	6.485
5.282		6.221	5.349	6.625	6.482		
4.834		5.723	5.933	6.177	6.869		

5.188	5.597	5.823	6.174	
5.386				
5.246733333	5.840387097		6.468451613	6.451115385
5.294	5.885		6.476	6.476
447.8	497.8		547.8	547.8
± 0.885	± 0.6345		± 0.564	± 0.357

Table 5: Measurements for data sets 3-2, 3-3, and 4-3

3-2		3-3		4-3	
155.766	156	189.667	191.167	162.675	163.672
154.253	154.001	192.667	191.707	162.003	160.502
154.002	157.851	195.668	185.783	162.5	163.773
155.312		191.11		162.521	
0.966	1.009	1.042	1.055	0.985	0.964
1.073	1.095	0.985	0.977	1.077	0.995
0.987		1.008		1.088	
Left edge	Right Edge	Left edge	Right Edge	Left edge	Right Edge
7.041	7.061	7.405	7.169	10.8	10.522
6.955	6.718	7.98	7.352	10.77	10.4
7.17	6.954	7.884	7.68	10.43	10.774
6.978	6.933	7.404	7.536	10.557	10.251
7.212	6.89	7.771	7.745	10.863	10.896
6.89	7.064	7.692	7.378	10.687	10.493
		7.536	7.956		10.585
			7.329		10.773
6.763	6.665	7.433	7.535	10.155	10.032
7.211	7.179	7.902	7.222	10.739	10.465
6.826	7.471	7.43	7.536	10.28	9.97
7.115	7.212	7.85	7.589	10.834	10.374
	data cropped out			Doesn't Exist	
7.052	6.632	7.589	7.614		
6.89	6.954	7.3	7.955		
6.993166667		7.584148148		10.52954545	
7.067		7.658		10.023	
597.8		647.8		847.8	
± 0.4195		± 0.4055		± 0.463	

Table 6: Short-range measurements at 100 cm, 80 cm, and 60 cm

100cm	80cm	60cm
Chose to start with 100cm for mentality	too consistent and closeby additional measurement is deemed unnecessary	

146.503	145.255	199.167	199.379	226.002	226.003
145.333	145.668	199.506	199.878	225.938	226.256
146.51	146.168	199.357		226.168	226.295
145.906		199.4574		226.11	
Just as good as I needed it for this distance :D					
1.017	1.005	1.019	1.036	0.984	1.021
1	1.057	1.019	1.028	0.999	1.058
1.028		1.103		1.028	
One error could be middle ones reach max exposure on their edge and the fringes on the edge don't, but I measure the point of exposure change to max . This could also effect as we go out further	One error could be middle ones reach max exposure on their edge and the fringes on the edge don't, but I measure the point of exposure change to max . This could also effect as we go out further		last one not measured	Very unclear	
Left Edge	Right edge	Left Edge	Right edge	Left Edge	Right edge
1.154	1.243	0.919	0.928	0.631	0.759
1.166	1.234	0.898	0.909	0.708	0.66
1.217	1.14	0.947	0.947	0.752	0.687
1.243	1.26	0.997	0.996	0.844	0.759
1.123	1.165	0.953	0.943	0.745	0.8
1.218	1.199	0.94	0.914	0.734	0.696
		0.915			
Very unclear					
1.062	1.115	0.953	0.821	0.756	0.747
1.303	1.15	1.028	0.953	0.708	0.763
1.097	1.226	1.009	1.028	0.807	0.636
1.08	1.131	0.984	0.936	0.829	0.73
1.1763		0.94847619		0.73755	
1.182222		0.945778		0.709	
100		80		60	
±0.1205		±0.1035		±0.422	

Table 7: Short-range measurements at 40 cm and 20 cm

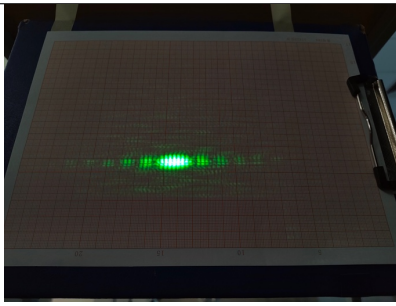
40cm		20cm	
208.417	208.5	232.417	231.75
207.667	208.375	232.855	232.629

208.625	209.188	231.5	231.878
208.462		232.172	
1.011	1.043	1.001	1.007
0.983	0.991	0.984	1.023
1.037		1.005	
Left edge	Right Edge	Left edge	Right Edge
0.496	0.424	0.319	0.287
0.468	0.512	0.287	0.266
0.552	0.552	0.276	0.291
0.488	0.504	0.259	0.226
0.512	0.5	0.255	0.291
0.476	0.552		
Able to see better with practice.	Able to see better with practice.	Able to see better with practice.	Able to see better with practice.
(Could've redone the beginning ones for better accuracy, but out of time limitations)	(Could've redone the beginning ones for better accuracy, but out of time limitations)	(Could've redone the beginning ones for better accuracy, but out of time limitations)	(Could've redone the beginning ones for better accuracy, but out of time limitations)
0.528	0.444	0.276	0.294
0.492	0.488	0.276	0.273
0.504	0.544	0.255	0.316
0.5	0.464	0.244	0.28
0.5		0.276166667	
0.473		0.236	
40		20	
± 0.064		± 0.0465	

4 Limitations and proposed improvements

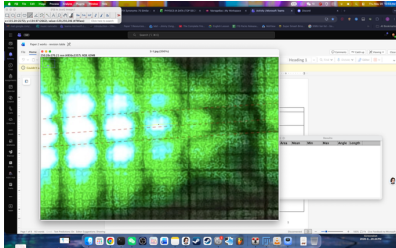
Table 8: Specific data issues, methodological weaknesses, and realistic improvements

Specific Data (quoted from “Data Table 1” etc) that shows lack of precision or inaccuracy	Likely weakness in methodology or limitation that caused this lack of precision or inaccuracy in data	Realistic (could be completed in our high school lab) explanation of improvement
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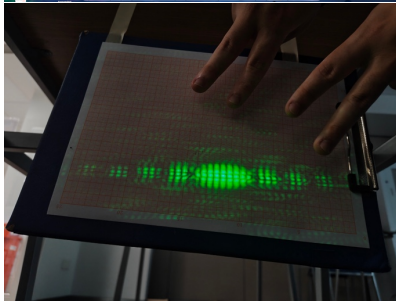
For the far/short-sided approach, I didn't tape the paper onto the clipboard

This could lead to distortions in the perspective correction process as it is not inherently flat



Some of the interference pattern isn't perfectly horizontal and are slanted

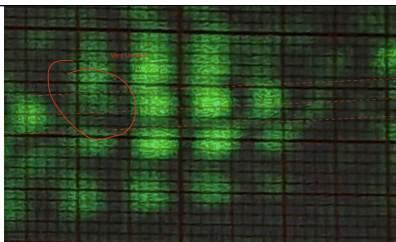
I had to draw lines that are parallel to the angle, and use that as a reference to accurately measure the distance in the software with greater validity



Some of the lines I have drawn don't perfectly line up throughout the horizontal component of the interference pattern

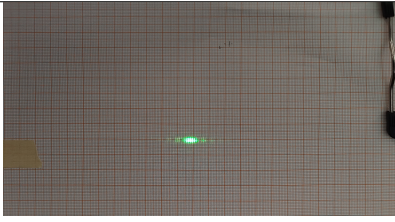
I genuinely don't know why =

Table 9: Additional sources of uncertainty



Some of the fringes are too unclear to be seen

I could spend more time adjusting camera properties and room lighting to ensure that the fringes are clear as well as the graphing paper's lines



The fringes close to the middle are usually brighter, so it would more easily reach the maximum exposure of the camera

This can affect the perceived “edge” is as that is my [baseline] for measuring fringe separation. I measure it by left/right edge to the edge of the adjacent fringe. This can compromise the data and introduce uncertainty. For example, when measuring distance between adjacent fringes of the leftmost edge, the left edge of that pair, would be more close to the center than the right fringe of the pair, since this one is more brighter, the edge would be further from the center of the fringe.
